

Question		Answer	Mark	Mark scheme	Additional guidance
24	(a)	8	P1	for process to find sum of unknown probabilities, eg $1 - 0.45 - 0.25 (= 0.3)$ OR to find the total number of counters in the bag, eg $\frac{18}{0.45} (= 40)$ OR to find the number of yellow counters, eg $\frac{0.25}{0.45} \times 18 (= 10)$	Award mark for any two probabilities given that sum to 0.3 eg given in the table.
			P1	for process to find $P(\text{red}) = 0.2$ oe or $P(\text{white}) = 0.1$ oe OR for process to find the total number of red and white counters, eg “40” – 18 – “10” (=12) OR for process to derive an equation in x, eg $2x + x = 1 - 0.45 - 0.25$ or $2x + x = \text{“0.3”}$ or $x = 0.1$	
			P1	for a complete process to find the number of red counters, eg $\frac{2 \times 0.1}{0.45} \times 18$ or $\frac{2}{3} \times \text{“12”}$ or $0.2 \times \text{“40”}$ or $\frac{0.2}{0.025}$	
			A1	cao	
	(b)	Explanation	C1	for explanation eg 0.5 multiplied by an odd number will never be a whole number, for half of a number to be an integer that number must be even, you can’t have half a marble	Award P2 for $P(\text{red})$ or $P(\text{white})$ (could be shown in table) Equations could be given as written statements or working but must be fully equivalent.